

bn240403498 - analysis report

bn240403498 — GRB 240403A

Analysis report — generated by the gated agentic pipeline.

- repository commit: f6c41b3
- products: results/sweep106/bn240403498
- P0 (frozen prediction) status: **none on file**
- All numbers are PROVISIONAL until the campaign is frozen and signed off.

1 · Identity and prior literature

No refereed literature found for this burst — our fits are the first published time-resolved analysis. (A burst is only ‘paperless’ after the recovery channels have run: dotted trigger form, local-PDF grep, citation graph, Google Scholar.)

2 · Data inventory and detector selection

detector	angle (°)	background pre / post (s)	source (s)	approval
n0	43.1	-83.2...-8.2 / 53.8...195.8	18.00...48.00	human_gui
n1	22.6	-86.9...-2.0 / 85.0...175.7	18.00...48.00	human_gui
n2	41.8	-86.9...-2.0 / 85.0...175.7	18.00...48.00	human_gui
n5	35.3	-86.9...-2.0 / 85.0...175.7	18.00...48.00	human_gui
b0	37.5	-86.9...-2.0 / 85.0...175.7	18.00...48.00	human_gui

Detector selection, background windows and the source interval are **human decisions adopted unchanged** by the pipeline, not re-derived. Individual detectors legitimately carry different background windows.

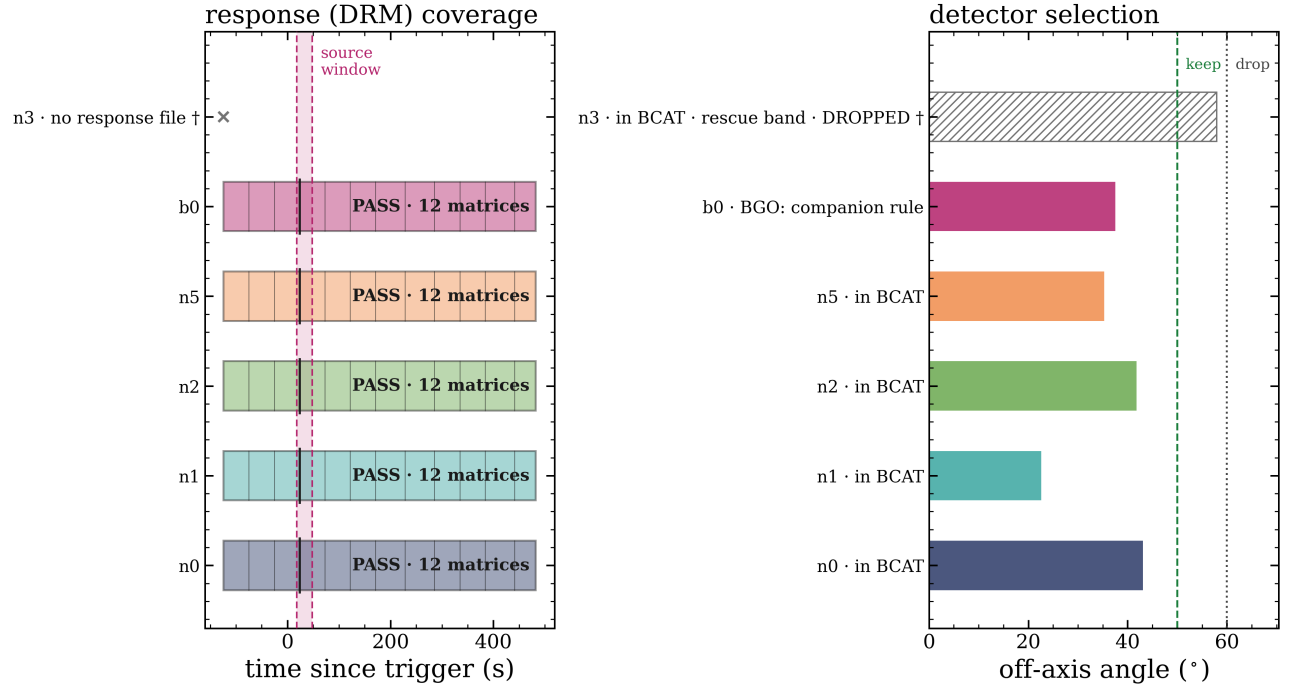
Step 1–2. Left: each approved detector’s response (DRM) coverage against the stamped source window — a bar must bracket the window for the effective area to be valid. Right: off-axis angles against the NaI selection rule — 50° keep; 50–60° kept only via the BCAT rescue (the detector triggered); >60° drop. BGOs are exempt (companion rule) and appear for completeness.

3 · Background and source intervals

Step 3. Per detector: the light curve (grey) with the fitted background polynomial (colour) through the approved pre/post windows (green); the source interval is red. The diagnostic question is whether the polynomial tracks the data inside the green windows AND extrapolates sanely beneath the burst.

Step 4. The source window (red) inside the common background gap (green). Where the source overruns the gap the figure says so — those are ledgered, accepted human overrides, not defects.

bn240403498 — 5 approved detectors, all DRM-valid over the source window; n3 in BCAT but dropped



PASS = DRM coverage brackets the stamped source window [18, 48] s; thin ticks = matrix boundaries
8 further detectors (81.5–145.1°, all beyond the axis) are not shown — all DROP: b1, n4, n6, n7, n8, n9, na, nb
no glg_* files on disk and LAT boresight 62° (GCN 36024): no LLE/LAT expected. Rails (NaI-only): green dashed = 50° keep, dotted = 60° drop; b0 is kept by the companion rule.
† n3 (57.94°) is exactly the written 50–60° BCAT rescue case: the WRITTEN RULE SAYS KEEP, the recorded human decision DROPPED it. A reason was elicited from the PI and recorded verbatim; the written rule is NOT amended and the flag is OPEN — see results/campaign/divergence_ledger.md. No n3 data were retrieved.
hatched row = in the BCAT mask but NOT approved.
thick tick: the window CROSSES a matrix boundary at 23.424 s — more than one DRM covers it
angles recomputed for this run from glg_poshist_all_240403_v00.fit — catalog DET_ANGLE reproduced exactly (max residual 0.00000°) via the repo angle code; an independent reimplement agrees to $\leq 0.01^\circ$

Figure 1: bn240403498_step1_inventory.png

bn240403498 — step 3: fitted background through the approved windows

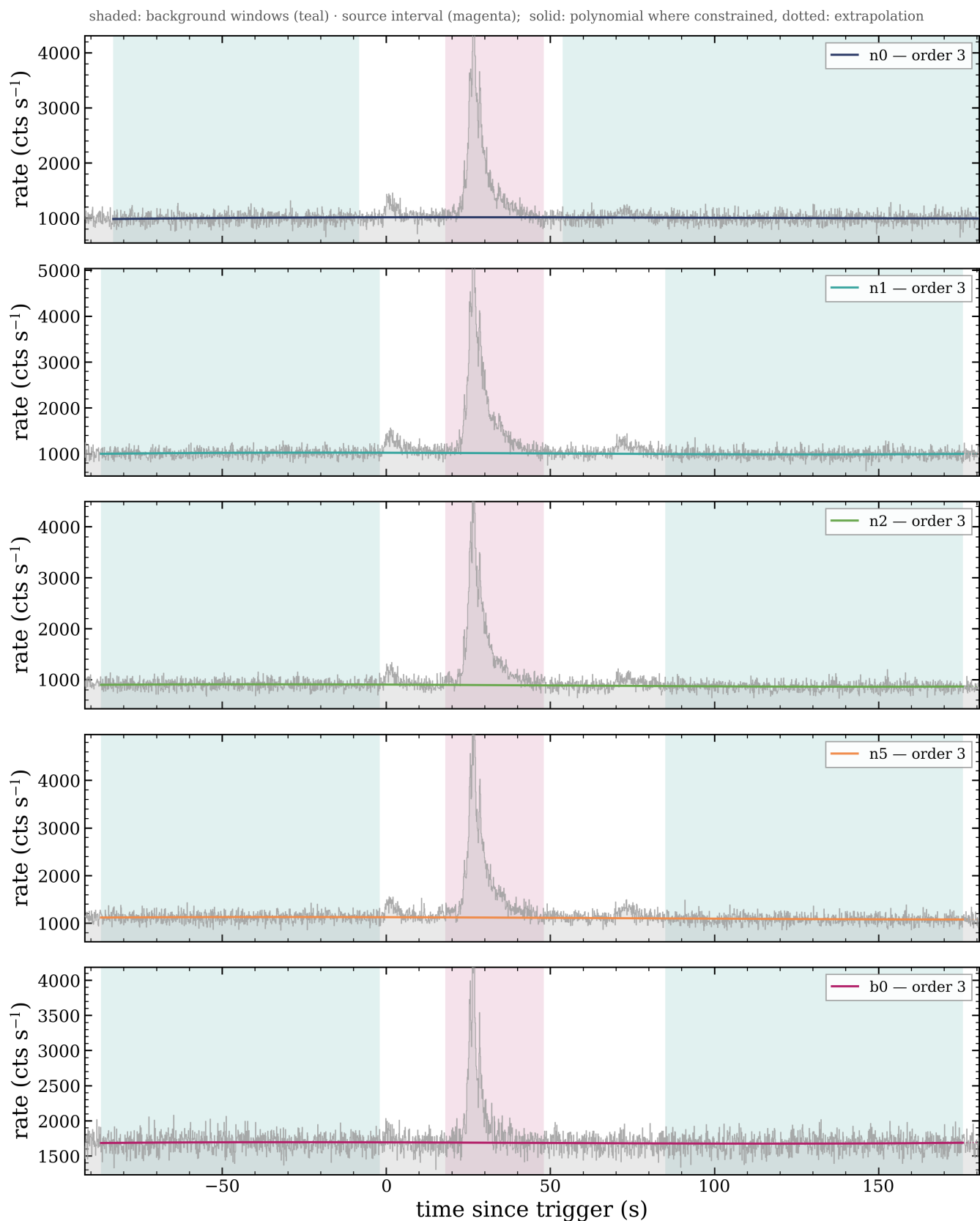


Figure 2: bn240403498_step3_background.png

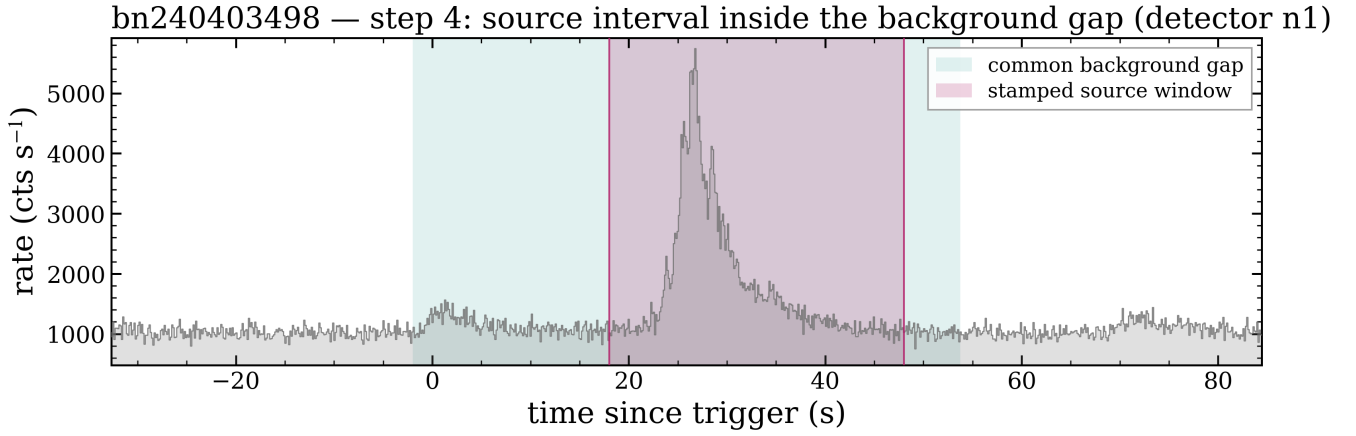


Figure 3: bn240403498_step4_source.png

4 · Time binning

15 blocks from Bayesian Blocks with a significance merge; per-block significance 5–73.

Narrowest 0.25 s, widest 4.29 s — the binning adapts to the light curve rather than using a fixed cadence.

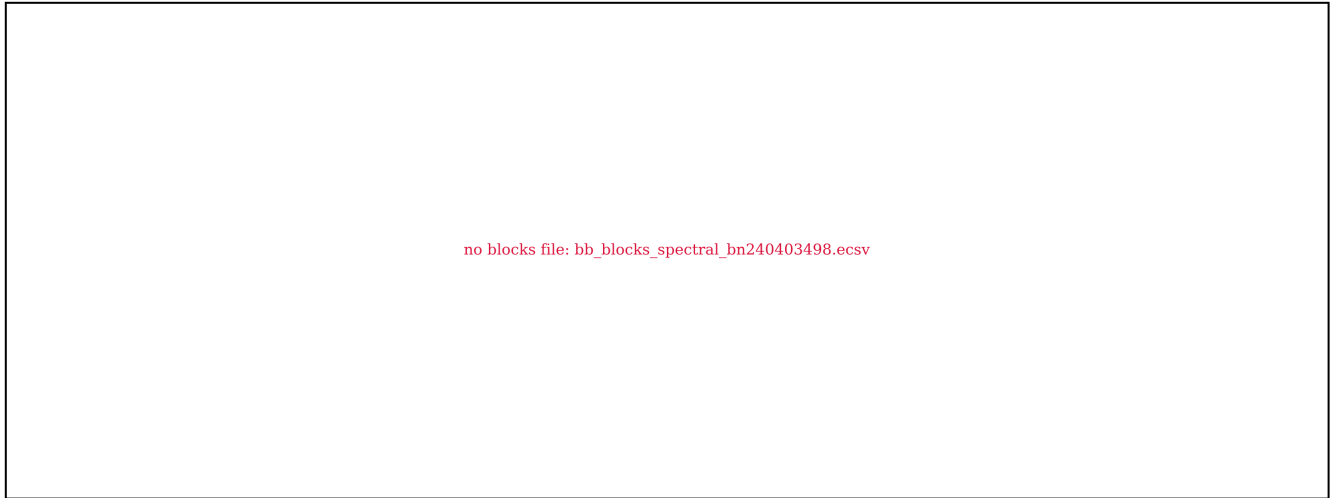


Figure 4: bn240403498_step5_binning.png

Step 5. Net light curve (filled histogram) with the Bayesian blocks drawn as a continuous step, coloured by block significance S . S is computed by threeML's own routine (Li & Ma equivalent for a Gaussian background, Vianello 2018) and is ACCUMULATED over the block, so it grows with duration as well as with rate: a long dim block can legitimately outscore a short bright one. Here the 2.83 s shoulder block scores above the 0.76 s peak block for exactly that reason.

5 · Time-resolved spectroscopy

Every block is fitted with the full model menu; the winner is the lowest AIC among **physically valid** fits, and evidence is reported as two margins (against the runner-up and against the best simple model).

block	interval (s)	winner	Δ AIC vs best simple	Band	Ep (keV)		thermal
T_INT	22.27...43.97	CPL+BB	DECISIVE	-1.06	329.7	-2.28	Band+BB kT=21.3 (in-band)
0	22.27...23.61	CPL+BB	+1.0	-1.28	191.7	-3.17	—
1	23.61...24.71	CPL	-2.4	-1.29	540.3	-9.37	—
2	24.71...25.21	CPL	-3.4	-0.93	457.8	-3.74	—
3	25.21...26.23	SBPL+BB	+0.8	-0.89	542.0	-2.78	—
4	26.23...27.04	Band+BB	+0.1	-0.84	573.5	-2.88	—
5	27.04...27.30	SBPL	-2.0	-0.76	248.9	-2.00	—
6	27.30...27.86	CPL+BB	STRONG	-0.70	178.5	-1.93	CPL+BB kT=25.1 (in-band)
7	27.86...28.31	SBPL	-0.7	-0.88	231.8	-2.01	—
8	28.31...28.82	Band	-2.0	-0.78	254.4	-2.12	—
9	28.82...29.40	SBPL	-0.2	-0.99	214.6	-2.08	—
10	29.40...30.69	SBPL	-1.2	-1.01	182.2	-2.09	—
11	30.69...32.47	Band	-0.5	-0.94	127.3	-2.29	—
12	32.47...35.84	CPL	-4.0	-1.08	114.7	-2.31	—
13	35.84...39.68	CPL	-3.0	-1.21	91.4	-2.59	—
14	39.68...43.97	CPL+BB	+0.1	-1.65	78.1	-8.71	—

Spectral evolution: Ep runs 192 → 78 keV across the burst (hard-to-soft).

figure missing: bn240403498/spectral_evolution.png — Step 6. Fitted parameter evolution across blocks. Discontinuities that no emission mechanism could produce are the signature of a collapsed fit, not of the source.

figure missing: bn240403498/ep_kt_correlation.png — Step 6. Peak energy against blackbody temperature where a thermal component is significant. Absent when no block carries one.

figure missing: bn240403498_nuFnu_best_montage.png — Step 8. F spectrum per block with the **engine's stored winning model** — the panel displays the stored fit, it does not re-decide. Any [! PANEL!=ENGINE] stamp means the drawn curve drifted from the table: trust the table.

6 · Temporal properties

- **T90 = 13.73 ± 0.36 s** (reference detector n1)
- T50 = 4.36 s
- minimum variability timescale: **withheld** — the catalog MVT_* columns are STALE-PENDING-REWALK and this burst is not in **rewalked_triggers**
- spectral lag: **withheld** — the catalog LAG_* columns carry the proven sign-inverted DCCF; the validated engine (s02c_spectral_lag, via scripts/47c) was not available for this burst

NR-31 guard active. Any MVT or lag for this burst must come from a re-walk with the validated tools, not from **temporal_catalog_all106.ecsv**.

Uncertainties come from Poisson realizations of the raw counts minus the fitted background, using the same estimator as the point value inside the approved window (audited 2026-08-13). Background-model uncertainty is not propagated.

Step 7. Cumulative net counts per energy band with t5/t95 marked. The duration shortening toward higher energy is the expected band dependence — measured here per burst rather than assumed from a population relation.

7 · Quality control and honest flags

Step 9. Top: per-block evidence for extra structure against the DECISIVE (Δ AIC = 10) and STRONG (= 6) lines, winner labelled. Bottom: 3.92 · kT for significant blackbodies against the 20 / 30 keV edge boundaries.

bn240403498 — step 7: energy-resolved T_{90} (detector n1); dashed = t_5 , t_{95}

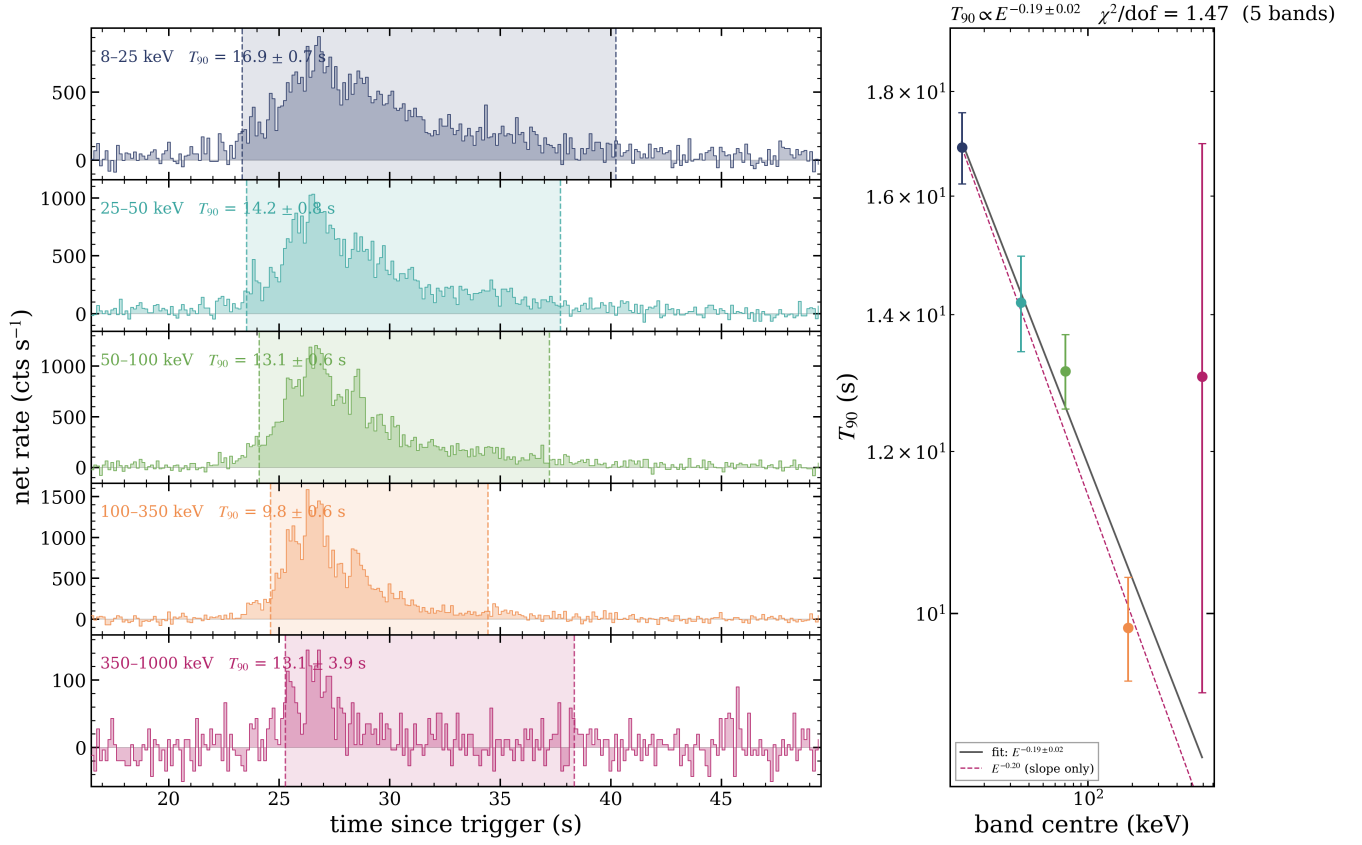


Figure 5: bn240403498_step7_temporal.png

no fit table: results/sweep106/bn240403498/bn240403498/spectral_fits.ecsv

Figure 6: bn240403498_step9_qc.png

No edge-constrained components, no near-threshold margins, no missing products.

8 · Provenance and standing caveats

- Stage-1 selections are human decisions, adopted unchanged and stamped.
- Fits are blind: the model menu and validity gates decide, never a published value.
- Winners are AIC-selected among valid fits; **adequacy is not identity** — a winning model is not a claim that the source is that model.
- The significant-blackbody rule currently counts the Band+BB and CPL+BB nested tests; other nested pairs are not yet included (open item), so thermal counts are **provisional**.
- All numbers here are provisional until the campaign is frozen and signed off.